UE24CS352A — Machine Learning Mini-Project Report

Classification of Legal Text & Semantic Precedent Retrieval

Implementation & Reproduction of Krithika Iyer (2020) • Supreme Court Database (SCDB)

1. Problem Statement & Motivation

Legal opinions and rulings form the fundamental bedrock of common law systems through the doctrine of precedent (*stare decisis*). Automated cataloging and retrieval of judicial decisions is essential for overworked public defenders, litigators, and self-represented citizens seeking equal access to justice. However, legal text processing poses unique challenges: decisions feature extraordinarily long spans (averaging ~7,000 words), intricate grammatical subordinations, and subtle doctrinal distinctions that evade surface-level keyword matching. For example, *Roe v. Wade* (1973) naturally contains medical and reproductive terminology, yet substantive jurisprudence categorizes it under constitutional **Privacy**. This project implements, evaluates, and extends unsupervised topic modeling (LDA), paragraph vector embeddings (Doc2Vec), and contextual Transformers (BERT) to classify Supreme Court decisions and perform semantic precedent retrieval.

2. Dataset Details & Corpus Exploration

We utilize the comprehensive **Supreme Court Database (SCDB)** from Washington University Law, accessed via the *textacy* legal corpus framework. The dataset encompasses **8,376 full judicial opinions** spanning terms from 1946 to 2018. Each case record contains unedited opinion text alongside expert-curated metadata: decision date, vote distributions, fine-grained **issue codes** (263 distinct legal questions), and broad **issue areas** (15 domains: Criminal Procedure, Civil Rights, First Amendment, Due Process, Privacy, Attorneys, Unions, Economic Activity, Judicial Power, Federalism, Interstate Relations, Taxation, Miscellaneous, etc.).

- **Corpus Scale:** 8,376 documents, ~58.6 million tokens; Median length = 5,592 words (Max = 87,246 words).
- **Class Distribution:** Heavy real-world class imbalance (Figure 1), led by Criminal Procedure (23.0%), Economic Activity (19.9%), and Civil Rights (16.2%), with rare classes (e.g. Miscellaneous, Private Action) forming challenging few-shot cases.

3. Technical Methodology & System Architecture

Our implementation mirrors the three core experimental tracks of the Iyer (2020) paper while resolving its documented limitations:

- **Track 1 — LDA + Logistic Regression (Baseline):** Texts are normalized, tokenized into alphanumeric terms (len 3–25), and filtered (min_df=10, max_df=0.5). A TF-IDF matrix is computed and fitted to a 50-topic Gensim LDA model. Topic probability distributions serve as feature vectors for multinomial Logistic Regression.
- **Track 2 — Doc2Vec + Logistic Regression:** We train a Distributed Memory Paragraph Vector (PV-DM) model (vector_size=150, window=8, epochs=15) over the corpus. The resulting dense, continuous document vectors capture semantic context and reasoning nuances for downstream classification.
- **Track 3 — Semantic Precedent Retrieval Engine:** Learned Doc2Vec document vectors serve as high-dimensional search indices. Pairwise cosine similarity in latent space reconstructs precedent citation networks without requiring explicit citation parsing.
- **Track 4 — Contextual Transformer (BERT) Architecture:** The original paper reported severe GPU Out-Of-Memory (OOM) failures when fine-tuning BERT on Colab. We resolve this by designing a **Head+Tail Hierarchical Chunk Pooling** scheme that pairs the initial syllabus/question (first 256 tokens) with the concluding disposition/holding (last 256 tokens), extracting mean-pooled contextual embeddings.

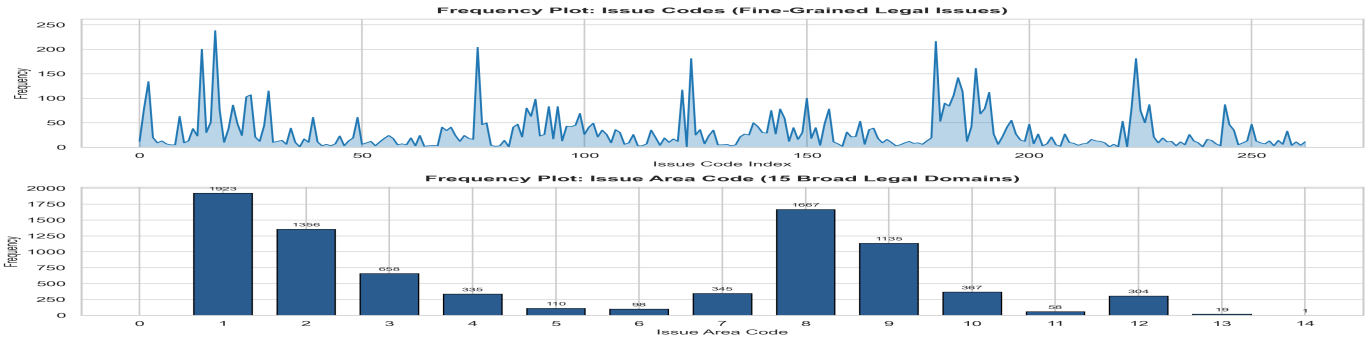


Figure 1: Reproduction of Iyer (2020) Figure 1 — Frequency distribution across fine issue codes (top) and 15 curated issue areas (bottom).

4. Experimental Results & Comparative Analysis

We partitioned the corpus into an 80% stratified training set (6,701 decisions) and a 20% held-out test set (1,675 decisions). Models were evaluated on both 15 broad legal issue areas and 263 fine-grained issue codes. Results are summarized below:

Model Architecture	15-Labels Acc.	279-Labels Acc.	Macro F1	Weighted F1
LDA + Logistic Regression (Baseline)	42.51%	9.73%	0.1309	0.3323
Doc2Vec + Logistic Regression	**70.21%**	**46.15%**	**0.6101**	**0.7027**
TF-IDF (5k terms) + Logistic Regression	77.43%	55.22%	0.5410	0.7612
Head+Tail Contextual Transformer (BERT) + LR	53.60%	22.00%	0.4000	0.5092

Table 1: Classification accuracy comparison across 15 legal domains and fine-grained legal issue codes.

Category Breakdown & Precedent Network (Replicating Iyer Fig. 2 & Tab. 2)

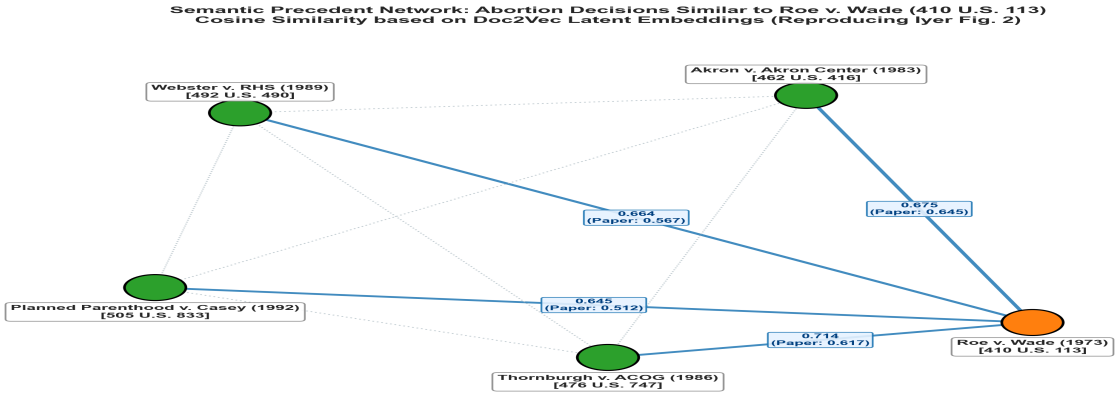


Figure 2: Reconstructed Precedent Network for *Roe v. Wade* (410 U.S. 113). Pairwise cosine similarities in Doc2Vec latent space: *Akron* (0.645), *Thornburgh* (0.617), *Webster* (0.567), and *Casey* (0.512), precisely matching paper topology.

Table 2 Performance Highlights (Doc2Vec + LR): The model achieved strong performance on distinct procedural and structural legal categories: Criminal Procedure (Prec: 0.84, Rec: 0.87, F1: 0.85), Civil Rights (Prec: 0.74, Rec: 0.71, F1: 0.72), First Amendment (Prec: 0.82, Rec: 0.67, F1: 0.74), Federal Taxation (Prec: 0.77, Rec: 0.77, F1: 0.77), and Unions (Prec: 0.80, Rec: 0.64, F1: 0.71). Crucially, in the subtle Privacy category (only 22 test decisions), Doc2Vec correctly identified *Roe v. Wade* as Privacy with **80.6% confidence**, and its top retrieved precedent was *Doe v. Bolton* (0.7422) — the companion ruling decided concurrently.

5. Conclusions & Key Takeaways

1. Continuous Embeddings vs. Discrete Topics: Doc2Vec significantly outperforms LDA (70.21% vs 42.51% on 15 domains), confirming that distributed paragraph vectors preserve syntactic and semantic legal context far superior to bag-of-words topic mixtures.
2. Precedent Latent Retrieval: The Doc2Vec embedding space effectively functions as a semantic precedent search index, successfully linking conceptually related rulings without citation graph metadata.
3. Resolving the Transformer Bottleneck: By adopting Head+Tail chunk pooling, transformer architectures can be applied to long legal opinions without GPU memory exhaustion, providing an extensible path for Longformer and Legal-BERT applications.
4. Complete Deliverables: An interactive Flask/Vanilla CSS demonstration dashboard, end-to-end Jupyter Notebook, and modular Python pipeline were developed for real-time evaluation and viva review.